BUSFIN 8200: Asset Pricing Foundations
Problem Set 2

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Question 1

1(a)

> **Task.** Do we have Complete Markets in this economy?

Yes.

Definition (Complete Markets). Markets are complete if there is one contingent claim (Arrow-Debreu security) for each state of nature $n = 1, \ldots, N$, where the contingent claim for state n pays \$1 if state n materializes and \$0 otherwise (Module 2, slide 1.3).

Let's define the payoff matrix

$$\mathbf{X} = \begin{bmatrix} 10 & 20 & 30 \\ 10 & 10 & 10 \\ 0 & 10 & 10 \end{bmatrix}$$

(rows: assets A, B, C; columns: states $n = 1, 2, 3$). Then there exists $\mathbf{\Omega}$ such that $\mathbf{X}'\mathbf{\Omega} = \mathbf{I}$, with $\mathbf{\Omega} = (\mathbf{X}')^{-1}$, if $\mathbf{X}'$ is invertible.

Even if a contingent claim is not traded directly, it exists in this economy if a portfolio of the traded assets A, B, and C replicates it, i.e., pays \$1 in state n and \$0 otherwise. $\mathbf{\Omega}$ tells us how many units of each asset to hold to replicate the contingent claims (column n of $\mathbf{\Omega}$ is the replicating portfolio for the contingent claim of state n). Since $\mathbf{X}'\mathbf{\Omega} = \mathbf{I}$, if $\mathbf{X}'$ is invertible, then a replicating portfolio for each contingent claim exists in this economy.

Therefore, checking whether $\mathbf{X}'$ is invertible is sufficient for Complete Markets in this economy. Since $\det(\mathbf{X}') = 1000 \neq 0$, $\mathbf{X}'$ is invertible. Therefore, this market is complete.

1(b)

> **Task.** Suppose that the Law of One Price (LOP) holds in this economy. Calculate the price of each Contingent Claim (or Arrow-Debreu security) in this economy. [Footnote 2: [...] You need to use the LOP to solve for $\mathbf{P}_c$ as a function of $\mathbf{X}$ and $\mathbf{P}$.]

Setup. Let $P_c(n)$ denote the price of the contingent claim for state n. Then

$$P(A) = \sum_{n=1}^{3} P_c(n)\,\text{Payoff}_A(n),$$
$$P(B) = \sum_{n=1}^{3} P_c(n)\,\text{Payoff}_B(n),$$
$$P(C) = \sum_{n=1}^{3} P_c(n)\,\text{Payoff}_C(n).$$

Method. Or, in matrix form,

$$\mathbf{P} = \mathbf{X}\mathbf{P}_c,$$

where $\mathbf{X}$ is the payoff matrix from Question 1(a),

$$\mathbf{P}_c = \big(P_c(1), P_c(2), P_c(3)\big)', \quad \text{and} \quad \mathbf{P} = \big(P(A), P(B), P(C)\big)' = (16, 8, 6)'.$$

Result. We confirmed in Question 1(a) that $\mathbf{X}$ is invertible. So, we can calculate $\mathbf{P}_c = \mathbf{X}^{-1}\mathbf{P}$. Therefore, the prices of the Contingent Claims are

$$\mathbf{P}_c = \mathbf{X}^{-1}\mathbf{P} = \begin{bmatrix} 0 & 0.1 & -0.1 \\ -0.1 & 0.1 & 0.2 \\ 0.1 & -0.1 & -0.1 \end{bmatrix} \begin{bmatrix} 16 \\ 8 \\ 6 \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.4 \\ 0.2 \end{bmatrix},$$

i.e., $P_c(1) = 0.2$, $P_c(2) = 0.4$, and $P_c(3) = 0.2$, where

$$\mathbf{X}^{-1} = \begin{bmatrix} 0 & 0.1 & -0.1 \\ -0.1 & 0.1 & 0.2 \\ 0.1 & -0.1 & -0.1 \end{bmatrix} \qquad (\det(\mathbf{X}) = 1000)$$

(rows: states $n = 1, 2, 3$; columns: assets A, B, C).

1(c)

> **Task.** Suppose that the LOP holds in this economy and introduce a new asset (referred to as asset D) with payoff vector $\text{Payoff}(D) = (30, 20, 10)$ for our three states of nature $n = (1, 2, 3)$. Calculate the price of asset D in this economy

Method. As calculated in Question 1(b), $\mathbf{P}_c = (0.2, 0.4, 0.2)'$. Taking the inner product of $\mathbf{Payoff}(D) = (30, 20, 10)'$ with $\mathbf{P}_c$ gives the price of asset D in this economy:

Result.

$$P(D) = \mathbf{Payoff}(D)'\,\mathbf{P}_c = \begin{bmatrix} 30 & 20 & 10 \end{bmatrix} \begin{bmatrix} 0.2 \\ 0.4 \\ 0.2 \end{bmatrix}$$
$$= 30 \cdot 0.2 + 20 \cdot 0.4 + 10 \cdot 0.2 = 16.$$

1(d)

Task. (1) Calculate the Stochastic Discount Factor (SDF) realization under each of the three states of nature in this economy.

(2) Does this economy satisfy No-Arbitrage (explain)?

(3) Is there anything interesting about the SDF in this economy (if so, explain what it means in economy terms)?

Setup. The SDF M satisfies $\mathbb{E}[M \cdot \text{Payoff}] = \text{Price}$ for every asset in this economy:

$$P(A) = \sum_{n=1}^{3} \pi(n)\, M(n)\, \text{Payoff}_A(n),$$
$$P(B) = \sum_{n=1}^{3} \pi(n)\, M(n)\, \text{Payoff}_B(n),$$
$$P(C) = \sum_{n=1}^{3} \pi(n)\, M(n)\, \text{Payoff}_C(n).$$

Method. Or, in matrix form,

$$\mathbf{P} = \mathbf{X}_\pi \mathbf{M}, \qquad \mathbf{X}_\pi = \begin{bmatrix} 10 \cdot \frac{1}{4} & 20 \cdot \frac{1}{2} & 30 \cdot \frac{1}{4} \\ 10 \cdot \frac{1}{4} & 10 \cdot \frac{1}{2} & 10 \cdot \frac{1}{4} \\ 0 & 10 \cdot \frac{1}{2} & 10 \cdot \frac{1}{4} \end{bmatrix}$$

(rows: assets A, B, C; columns: states $n = 1, 2, 3$; $\mathbf{X}$ with column n multiplied by $\pi(n)$), where $\mathbf{M} = \big(M(1), M(2), M(3)\big)'$ and $\mathbf{P} = \big(P(A), P(B), P(C)\big)'$.

Result (1). Since $\det(\mathbf{X}_\pi) = 31.25 \neq 0$, $\mathbf{X}_\pi$ is invertible, with

$$\mathbf{X}_\pi^{-1} = \begin{bmatrix} 0 & 0.4 & -0.4 \\ -0.2 & 0.2 & 0.4 \\ 0.4 & -0.4 & -0.4 \end{bmatrix}$$

(rows: states $n = 1, 2, 3$; columns: assets A, B, C). Therefore, the realizations of the SDF are

$$\mathbf{M} = \mathbf{X}_\pi^{-1}\mathbf{P} = \begin{bmatrix} 0 & 0.4 & -0.4 \\ -0.2 & 0.2 & 0.4 \\ 0.4 & -0.4 & -0.4 \end{bmatrix} \begin{bmatrix} 16 \\ 8 \\ 6 \end{bmatrix} = \begin{bmatrix} 0.8 \\ 0.8 \\ 0.8 \end{bmatrix},$$

i.e., $M(1) = M(2) = M(3) = 0.8$.

Interpretation (2). This economy satisfies No-Arbitrage.

Definition (No-Arbitrage). (Module 2, slide 1.3)

- An asset with Payoff ≥ 0 must have Price ≥ 0.
- An asset with Payoff ≥ 0 and a chance for Payoff > 0 must have Price > 0.

Each Arrow-Debreu contingent claim has Payoff ≥ 0 with a chance for Payoff > 0, and from Question 1(b) its price is positive ($P_c(1) = 0.2$, $P_c(2) = 0.4$, $P_c(3) = 0.2$). The contingent claims therefore satisfy the No-Arbitrage condition above. Moreover, with complete markets and the LOP, the price of any payoff is determined by

$$P = \sum_{n=1}^{3} P_c(n)\,\text{Payoff}(n)$$

(the formula used in Questions 1(b) and 1(c)). Since $P_c(n) > 0$ for all n, any asset with Payoff ≥ 0 has Price ≥ 0, and if its payoff is positive in at least one state, its price is positive. Hence No-Arbitrage holds in this economy.

Interpretation (3). As we can see, the realized SDF is the same in every state of the economy ($M(n) = 0.8$ for all n). Since M is constant across states, its covariance with every asset's return is zero, $\text{Cov}(M, R_j) = 0$. That is, this is an economy in which the expected excess return of every asset is zero (no risk premium).

1(e)

> **Task.** Show that the price that the SDF assigns to asset D in this economy is the same as the price that you calculated in Question (1c)

Method. Using the SDF from Question 1(d) ($M(n) = 0.8$ for all n), with $\text{Payoff}_D(1) = 30$, $\text{Payoff}_D(2) = 20$, and $\text{Payoff}_D(3) = 10$,

Result.

$$\begin{aligned} P(D) &= \sum_{n=1}^{3} \pi(n)\, M(n)\, \text{Payoff}_D(n) \\ &= 0.25 \cdot 0.8 \cdot 30 + 0.5 \cdot 0.8 \cdot 20 + 0.25 \cdot 0.8 \cdot 10 \\ &= 16, \end{aligned}$$

which is the same as the price of asset D calculated in Question 1(c).

Question 2

2(a)

> **Task.** (1) Find the SDF $M_t^* = a - \mathbf{b}'\mathbf{xR}_t = a - xR_t^*$ that unconditionally prices the excess returns of the 10 portfolios in the BM Dataset (ex-post). That is, find the SDF that satisfies $\widehat{\mathbb{E}}[M^* \cdot \mathbf{xR}] = \mathbf{0}$ for the $\mathbf{xR}$ vector in the BM Dataset. [Footnote 3: Let's use the normalization $\mathbb{E}[M^*] = 1$ (throughout Question 2). [...]]
>
> (2) Provide a scatterplot with the portfolio deciles (1 to 10) in the x-axis and the respective b coefficients in the y-axis.
>
> (3) Describe the pattern you observe in the scatterplot and explain why this pattern emerges.

Setup. The SDF $M_t^* = a - \mathbf{b}'\mathbf{xR}_t$ prices the excess returns $\mathbf{xR}_t$ of the 10 value-weighted book-to-market decile portfolios ("Lo 10" to "Hi 10", Ken French data library), in excess of the Fama–French risk-free rate. Data: monthly returns (one-month horizon), July 1927 to August 2026 ($T = 1190$ months), in decimals.

Method. With the normalization $\widehat{\mathbb{E}}[M^*] = 1$, a and $\mathbf{b}$ solve the sample moment conditions

$$\begin{aligned} a - \mathbf{b}'\,\widehat{\mathbb{E}}[\mathbf{xR}] &= 1, \\ a\,\widehat{\mathbb{E}}[\mathbf{xR}] - \widehat{\mathbb{E}}[\mathbf{xR}\,\mathbf{xR}']\,\mathbf{b} &= \mathbf{0}, \end{aligned}$$

where $\widehat{\mathbb{E}}[\cdot]$ is the time average.

Result (1)–(2). Figure 1 plots $\hat{\mathbf{b}}$ by decile (estimates in `output/q2a_sdf_coefficients.csv`).

Interpretation (3). *[Describe the pattern you observe in the scatterplot.]*
[Explain why this pattern emerges.]

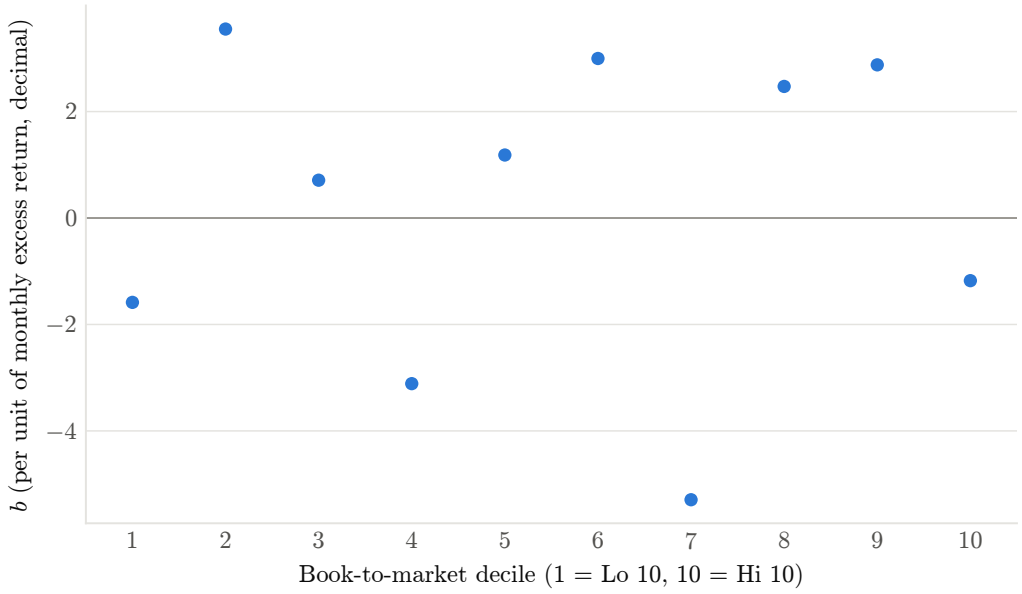


Figure 1: Coefficients $\hat{\mathbf{b}}$ of the SDF $M_t^* = a - \mathbf{b}'\mathbf{x}\mathbf{R}_t$ that prices the excess returns of the 10 value-weighted book-to-market decile portfolios, under the normalization $\widehat{\mathbb{E}}[M^*] = 1$. Data: monthly returns (one-month horizon), July 1927 to August 2026 ($T = 1190$ months); excess returns in decimals, so b is per unit of decimal monthly excess return.

2(b)

> **Task.** (1) Repeat Question (2a), but this time using the excess returns of the 10 portfolios in the MOM Dataset.
>
> (2) Calculate and report the correlation between the M_t^* from this question and the M_t^* from Question (2a).
>
> (3) Interpret this correlation in economic terms.

Setup. As in Question 2(a), with the 10 value-weighted momentum decile portfolios ("Lo PRIOR" to "Hi PRIOR", prior 12–2 month return) as test assets. Data: monthly returns (one-month horizon), July 1927 to August 2026 ($T = 1190$ months), in decimals.

Method. The same moment conditions as in Question 2(a). The correlation of the two SDF series is the sample (Pearson) correlation.

Result (1)–(2). Figure 2 plots $\hat{\mathbf{b}}$ by decile. The sample (Pearson) correlation between the monthly SDF series $M^*_{\text{BM},t}$ estimated in Question 2(a) and $M^*_{\text{MOM},t}$ estimated in Question 2(b), over July 1927 to August 2026 ($T = 1190$ months), is

$$\widehat{\text{Corr}}\left(M^*_{\text{BM},t},\, M^*_{\text{MOM},t}\right) = 0.355.$$

Interpretation (1), (3). *[Describe the pattern you observe in the scatterplot.]*
[Explain why this pattern emerges.]
[Interpret this correlation in economic terms.]

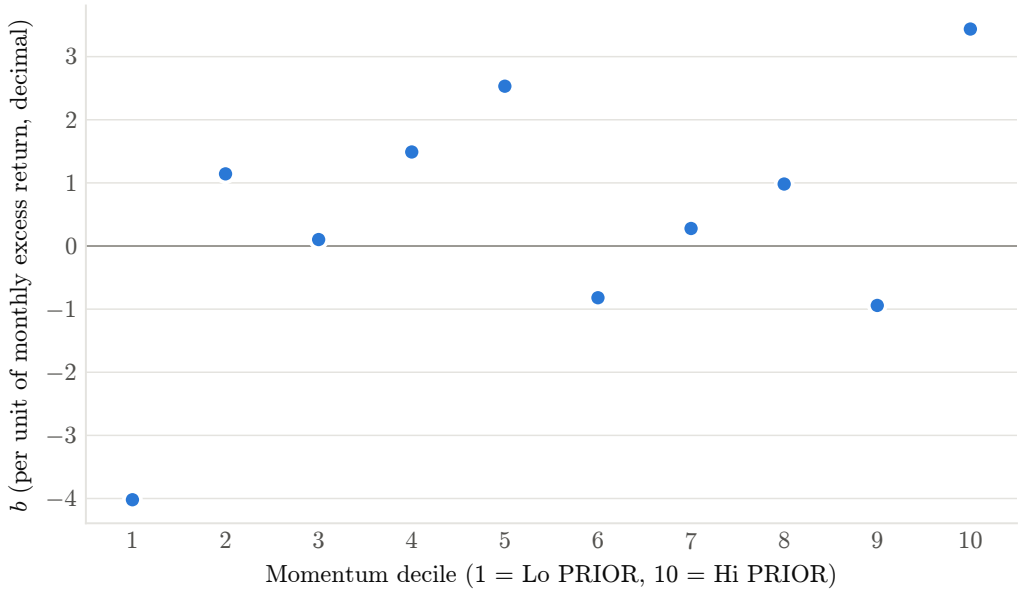


Figure 2: Coefficients $\hat{\mathbf{b}}$ of the SDF $M_t^* = a - \mathbf{b}'\mathbf{x}\mathbf{R}_t$ that prices the excess returns of the 10 value-weighted momentum (prior 12–2 month return) decile portfolios, under the normalization $\widehat{\mathbb{E}}[M^*] = 1$. Data: monthly returns (one-month horizon), July 1927 to August 2026 ($T = 1190$ months); excess returns in decimals, so b is per unit of decimal monthly excess return.

2(c)

> **Task.** (1) Repeat Question (2a), but this time using the excess returns of the 20 portfolios in the merge of the BM Dataset and the MOM Dataset. For the scatterplot, still have the portfolio deciles (1 to 10) in the x-axis, but use different colors for the dots associated with different sorting variables (blue for BM portfolios and red for MOM portfolios).
>
> (2) Calculate and report the Sharpe ratios of the xR_t^* portfolios obtained in Questions (2a), (2b) and (2c).
>
> (3) Explain what you observe and why it is (or it is not) in line with a key SDF property we learned about in the lectures.

Setup. As in Question 2(a), with the 20 portfolios of Questions 2(a) and 2(b) together as test assets. Data: monthly returns (one-month horizon), July 1927 to August 2026 ($T = 1190$ months), in decimals.

Method. The same moment conditions as in Question 2(a). For the SDFs of Questions 2(a)–2(c), the Sharpe ratio of $xR_t^* = \hat{\mathbf{b}}'\mathbf{x}\mathbf{R}_t$ is $\widehat{SR}(xR^*) = \widehat{\mathbb{E}}[xR_t^*]/\widehat{\sigma}(xR_t^*)$, with $\widehat{\sigma}$ dividing by T.

Result (1)–(2). Figure 3 plots $\hat{\mathbf{b}}$ by decile and Table 1 reports the Sharpe ratios.

Interpretation (1), (3). *[Describe the pattern you observe in the scatterplot.]*
[Explain why this pattern emerges.]
[Explain what you observe in the Sharpe ratios (Table 1) and why it is (or it is not) in line with a key SDF property from the lectures.]

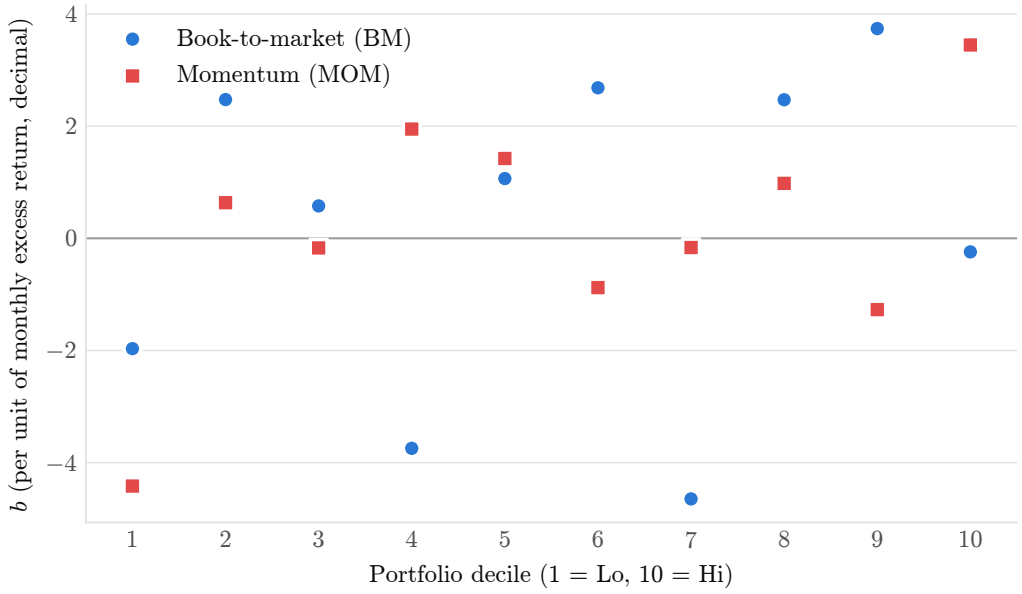


Figure 3: Coefficients $\hat{\mathbf{b}}$ of the SDF $M_t^* = a - \mathbf{b}'\mathbf{x}\mathbf{R}_t$ that prices the excess returns of the 20 value-weighted book-to-market (blue circles) and momentum (red squares) decile portfolios, under the normalization $\widehat{\mathbb{E}}[M^*] = 1$. Data: monthly returns (one-month horizon), July 1927 to August 2026 ($T = 1190$ months); excess returns in decimals, so b is per unit of decimal monthly excess return.

SDF	Test assets	$\widehat{SR}(xR^*)$
Question 2(a)	10 book-to-market deciles	0.192
Question 2(b)	10 momentum deciles	0.258
Question 2(c)	20 book-to-market and momentum deciles	0.305

Table 1: Sharpe ratios $\widehat{SR}(xR^*) = \widehat{\mathbb{E}}[xR_t^*]/\widehat{\sigma}(xR_t^*)$ of the portfolios $xR_t^* = \hat{\mathbf{b}}'\mathbf{x}\mathbf{R}_t$ implied by the SDFs of Questions 2(a)–2(c). Sample: July 1927 to August 2026 ($T = 1190$ months). Data: monthly returns (one-month horizon). Units: monthly Sharpe ratio, not annualized; $\widehat{\sigma}$ divides by T. No standard errors are reported.

2(d)

Task. (1) For each portfolio p (covering the 20 portfolios from Question (2c)), calculate (and report in a table) the pricing errors $\hat{\alpha}_p = \widehat{\mathbb{E}}[xR_p] - \widehat{\mathrm{Cov}}[-M^*, xR_p]$ under each M_t^* above (from Questions (2a), (2b), and (2c)).

(2) Describe the results and explain what we learn from them.

Setup. Pricing errors of the 20 portfolios of Question 2(c) under each of the three SDFs M_t^* estimated in Questions 2(a), 2(b) and 2(c). Data: monthly returns (one-month horizon), July 1927 to August 2026 ($T = 1190$ months).

Method. For each portfolio p and each SDF,

$$\hat{\alpha}_p = \widehat{\mathbb{E}}[xR_p] - \widehat{\mathrm{Cov}}[-M^*, xR_p],$$

$$\widehat{\mathrm{Cov}}[-M^*, xR_p] = \frac{1}{T}\sum_{t=1}^{T}(-M_t^* + \overline{M^*})(xR_{p,t} - \overline{xR}_p).$$

Result (1). Table 2.

Interpretation (2). *[Explanation will be here]*

	SDF M_t^* estimated in		
Portfolio	2(a) BM	2(b) MOM	2(c) BM + MOM
Panel A: book-to-market deciles			
1 (Lo 10)	0.000	−0.151	0.000
2	0.000	−0.010	0.000
3	0.000	−0.022	0.000
4	0.000	−0.078	0.000
5	0.000	0.050	0.000
6	0.000	0.126	0.000
7	0.000	0.028	0.000
8	0.000	0.238	0.000
9	0.000	0.345	0.000
10 (Hi 10)	0.000	0.320	0.000
Panel B: momentum deciles			
1 (Lo PRIOR)	−1.016	0.000	0.000
2	−0.505	0.000	0.000
3	−0.327	0.000	0.000
4	−0.149	0.000	0.000
5	−0.088	0.000	0.000
6	−0.026	0.000	0.000
7	0.051	0.000	0.000
8	0.190	0.000	0.000
9	0.224	0.000	0.000
10 (Hi PRIOR)	0.542	0.000	0.000

Table 2: Pricing errors $\hat{\alpha}_p = \widehat{\mathbb{E}}[xR_p] - \widehat{\mathrm{Cov}}[-M^*, xR_p]$ of the 20 book-to-market and momentum decile portfolios under the SDFs M_t^* estimated in Question 2(a) (10 book-to-market deciles), Question 2(b) (10 momentum deciles) and Question 2(c) (all 20 portfolios). Sample: July 1927 to August 2026 ($T = 1190$ months). Data: monthly value-weighted returns (one-month horizon) in excess of the Fama–French risk-free rate. Units: percent per month. Covariances divide by T. No standard errors are reported.

2(e)

Task. (1) Repeat the M^* construction in Question (2c), but using data until December 1989. Then, apply the $\hat{a}$ and $\hat{\mathbf{b}}$ obtained from this exercise to construct the out-of-sample SDF $M^*_{OS,t} = \hat{a} - \hat{\mathbf{b}}'\mathbf{x}\mathbf{R}_t$ from January 1990 to the end of the sample. For each portfolio p (covering the 20 portfolios from Question (2c)), calculate (and report in a table) the pricing error $\hat{\alpha}_{OS,p} = \widehat{\mathbb{E}}[xR_p] - \widehat{\mathrm{Cov}}[-M^*_{OS}, xR_p]$ using data from January 1990 to the end of the sample to estimate $\hat{\alpha}_{OS,p}$.

(2) Describe the results and explain what we learn from them.

Setup. Out-of-sample pricing errors of the 20 portfolios of Question 2(c). Data: monthly returns (one-month horizon); estimation sample July 1927 to December 1989 ($T = 750$ months), evaluation sample January 1990 to August 2026 ($T = 440$ months).

Method. $\hat{a}$ and $\hat{\mathbf{b}}$ are estimated as in Question 2(c) on the estimation sample. The out-of-sample SDF is $M^*_{OS,t} = \hat{a} - \hat{\mathbf{b}}'\mathbf{x}\mathbf{R}_t$ on the evaluation sample, and $\hat{\alpha}_{OS,p} = \widehat{\mathbb{E}}[xR_p] - \widehat{\mathrm{Cov}}[-M^*_{OS}, xR_p]$, with all moments computed on the evaluation sample and covariances dividing by T.

Result (1). Table 3.

Interpretation (2). *[Explanation will be here]*

Portfolio	$\hat{\alpha}_{OS,p}$
Panel A: book-to-market deciles	
1 (Lo 10)	1.097
2	0.786
3	0.900
4	0.696
5	0.799
6	0.718
7	0.698
8	0.592
9	0.827
10 (Hi 10)	0.971
Panel B: momentum deciles	
1 (Lo PRIOR)	2.705
2	1.648
3	1.390
4	1.208
5	0.918
6	0.830
7	0.760
8	0.644
9	0.432
10 (Hi PRIOR)	0.545

Table 3: Out-of-sample pricing errors $\hat{\alpha}_{OS,p} = \widehat{\mathbb{E}}[xR_p] - \widehat{\mathrm{Cov}}[-M^*_{OS}, xR_p]$ of the 20 book-to-market and momentum decile portfolios. $\hat{a}$ and $\hat{\mathbf{b}}$ are estimated as in Question 2(c) on July 1927 to December 1989 ($T = 750$ months); $M^*_{OS,t} = \hat{a} - \hat{\mathbf{b}}'\mathbf{x}\mathbf{R}_t$ and the moments are computed on January 1990 to August 2026 ($T = 440$ months). Data: monthly value-weighted returns (one-month horizon) in excess of the Fama–French risk-free rate. Units: percent per month. Covariances divide by T. No standard errors are reported.

Question 3

Sample periods. Results are reported for two sample versions, both ending in December 2021 (the last EQ observation). Version (1) "Common" uses December 1940 to December 2021 for every subquestion of Questions 3 and 4 (the latest start, set by z in Questions 3(c)–3(d)); version (2) "Max" uses the longest sample available for each subquestion. Version (1) puts Questions 3 and 4 on the same sample, so that differences between the non-linear SDF (Question 3, $M_t = R_{w,t}^{-\gamma}$) and its linearized version (Question 4, $M_t = a - b \cdot R_{w,t}$) are not driven by differences in the sample period; for example, Questions 3(e) and 4(b) use the same 21 moments. Version (2) uses all data available for each subquestion, so each estimate is as precise as the data allow.

Item	(1) Common	(2) Max
Question 3(b)	Dec 1940 – Dec 2021 ($T = 973$)	Dec 1927 – Dec 2021 ($T = 1129$)
Questions 3(c), 3(d)	Dec 1940 – Dec 2021 ($T = 973$)	Dec 1940 – Dec 2021 ($T = 973$)
Question 3(e)	Dec 1940 – Dec 2021 ($T = 973$)	Jun 1928 – Dec 2021 ($T = 1123$)
Questions 4(b)–4(e)	Dec 1940 – Dec 2021 ($T = 973$)	Jun 1928 – Dec 2021 ($T = 1123$)

Questions 3(c) and 3(d) are identical in both versions, so one set of results is reported.

3(a)

> **Task.** Using a power utility function, $U(W) = W^{1-\gamma}/(1-\gamma)$, with γ representing the constant relative risk aversion, show that the SDF $M_t = R_{w,t}^{-\gamma}$ prices the vector of excess returns (i.e., show that $\mathbb{E}_t[R_{w,t+1}^{-\gamma} \cdot \mathbf{xR}_{t+1}] = \mathbf{0}$).

Method. From the maximization problem in Equation (3.2),

$$\boldsymbol{\varpi}_t^* = \arg\max_{\boldsymbol{\varpi}_t} \mathbb{E}_t\big[U(W_{t+1})\big] \quad \text{s.t.} \quad W_{t+1} = W_t \cdot \big(R_{f,t+1} + \boldsymbol{\varpi}_t' \mathbf{xR}_{t+1}\big),$$

where $R_{w,t+1} = R_{f,t+1} + \boldsymbol{\varpi}_t^{*\prime} \mathbf{xR}_{t+1}$ is the return on the optimal portfolio, so that $W_{t+1} = W_t \cdot R_{w,t+1}$.

The first-order condition is

$$\begin{aligned} & \frac{\partial\, \mathbb{E}_t[U(W_{t+1})]}{\partial \boldsymbol{\varpi}_t} = \mathbb{E}_t\big[U'(W_{t+1}) \cdot W_t \cdot \mathbf{xR}_{t+1}\big] = \mathbf{0} \\ \iff \quad & \mathbb{E}_t\big[U'(W_{t+1}) \cdot \mathbf{xR}_{t+1}\big] = \mathbf{0}. \end{aligned}$$

Since W_t is the investor's wealth at time t, it is positive ($W_t > 0$) and known at time t, so it can be taken outside $\mathbb{E}_t[\cdot]$ and divided out.

Because $U'(W_{t+1}) = W_{t+1}^{-\gamma} = (W_t \cdot R_{w,t+1})^{-\gamma}$,

$$\begin{aligned} & \mathbb{E}_t\big[(W_t \cdot R_{w,t+1})^{-\gamma} \cdot \mathbf{xR}_{t+1}\big] = \mathbf{0} \\ \iff \quad & W_t^{-\gamma}\, \mathbb{E}_t\big[R_{w,t+1}^{-\gamma} \cdot \mathbf{xR}_{t+1}\big] = \mathbf{0} \\ \iff \quad & \mathbb{E}_t\big[R_{w,t+1}^{-\gamma} \cdot \mathbf{xR}_{t+1}\big] = \mathbf{0}, \end{aligned}$$

where the last step uses the same argument for $W_t^{-\gamma} > 0$.

Result. Therefore, $M_{t+1} = R_{w,t+1}^{-\gamma}$ prices the vector of excess returns. Because an SDF that prices only excess returns still does so after multiplication by any positive factor (here, $W_t^{-\gamma}$), the scale of M is not identified; a normalization is therefore needed and is innocuous.

3(b)

Task. (1) Estimate γ by GMM using the unconditional SDF equation $\mathbb{E}[R_{w,t}^{-\gamma} \cdot xR_{w,t}] = 0$. Use the proxies $R_{w,t} = R_{e,t} = e^{r_{e,t}}$ and $R_{f,t} = e^{r_{f,t}}$, with r_e and r_f from the EQ Dataset. [Footnote 5: [...] you still need to use the GMM minimization problem $\hat{\gamma} = \arg\min \widehat{\mathbb{E}}[R_{w,t}^{-\gamma} \cdot xR_{w,t}]$. This minimization problem implies that the GMM estimate solves $\hat{G} \cdot \widehat{\mathbb{E}}[R_{w,t}^{-\hat{\gamma}} \cdot xR_{w,t}] = 0$ where $\hat{G} = \widehat{\mathbb{E}}[\partial_\gamma (R_{w,t}^{-\hat{\gamma}} \cdot xR_{w,t})]$.]

(2) Report the γ estimate as well as its standard error. When estimating $\text{aVar}[\hat{\gamma}]$, estimate S using Newey and West (1987, 1994).

Setup. GMM estimate of γ from the unconditional SDF equation $\mathbb{E}[R_{w,t}^{-\gamma} \cdot xR_{w,t}] = 0$, with $R_{w,t} = e^{r_{e,t}}$ and $xR_{w,t} = e^{r_{e,t}} - e^{r_{f,t}}$ from the EQ Dataset (annual returns observed monthly). Samples: (1) Common, December 1940 to December 2021 ($T = 973$); (2) Max, December 1927 to December 2021 ($T = 1129$).

Method. With one moment and one parameter, $\hat{\gamma}$ solves $\hat{G} \cdot \widehat{\mathbb{E}}[R_{w,t}^{-\hat{\gamma}} \cdot xR_{w,t}] = 0$ with the analytical $\hat{G} = \widehat{\mathbb{E}}[\partial_\gamma (R_{w,t}^{-\gamma} \cdot xR_{w,t})]$ (footnote 5), and $\text{aVar}[\hat{\gamma}] = \hat{S}/(T\hat{G}^2)$ with $\hat{S}$ the Newey–West estimate.

Result (1)–(2). Table 4.

	(1) Common	(2) Max
$\hat{\gamma}$	2.8878	1.9061
	(0.7051)	(0.5930)
Newey–West lag	22	23
Sample	Dec 1940 – Dec 2021	Dec 1927 – Dec 2021
T (months)	973	1129

Table 4: GMM estimate of γ from $\widehat{\mathbb{E}}[R_{w,t}^{-\gamma} \cdot xR_{w,t}] = 0$, with $R_{w,t} = e^{r_{e,t}}$ and $xR_{w,t} = e^{r_{e,t}} - e^{r_{f,t}}$ from the EQ Dataset, one column per sample version. Samples: (1) Common: December 1940 to December 2021 ($T = 973$ months); (2) Max: December 1927 to December 2021 ($T = 1129$ months). Data: monthly observations of 12-month (annual) real returns. Units: decimals. Standard errors in parentheses below the estimates: Newey–West (1987, 1994), Bartlett kernel, Newey–West (1994) automatic lag rounded down (selected lag in the table).

3(c)

Task. (1) Estimate γ by GMM using the unconditional SDF equations $\mathbb{E}[R_{w,t}^{-\gamma} \cdot xR_{w,t}] = 0$ and $\mathbb{E}[R_{w,t}^{-\gamma} \cdot (z_{t-1} \cdot xR_{w,t})] = 0$. The second equation reflects the unconditional pricing of a managed portfolio with $z_t = \mathbb{E}_t^{OS}[xR_e]/\sigma_e^2$ as the trading instrument (and it captures a conditional implication of the model). [Footnote 6: σ_e^2 should be set to the unconditional variance of $R_{e,t}$ over the sample while $\mathbb{E}_t^{OS}[xR_e]$ is the out-of-sample forecast of $xR_{e,t+1}$ obtained from Question (2e) of Problem Set 1. [...]] Use an identity weighting matrix when minimizing pricing errors (so that the GMM estimate implicitly uses $\hat{\mathbf{A}} = \hat{\mathbf{G}}'$).

(2) Produce a table that reports the γ estimate as well as its standard error. The table should also report the value and t-statistic for the pricing error of each moment condition as well as the p-value for a χ^2 test with the null hypothesis that the pricing errors are jointly zero. When estimating $\text{aVar}[\hat{\gamma}]$ and $\text{aVar}[\widehat{\mathbb{E}}[\epsilon(\hat{\gamma})]]$, estimate S using Newey and West (1987, 1994).

Setup. GMM estimate of γ from the two moment conditions $\mathbb{E}[R_{w,t}^{-\gamma} \cdot xR_{w,t}] = 0$ and $\mathbb{E}[R_{w,t}^{-\gamma} \cdot (z_{t-1} \cdot xR_{w,t})] = 0$, with $R_{w,t}$ and $xR_{w,t}$ as in Question 3(b). The instrument is $z_{t-1} = \widehat{\mathbb{E}}^{OS}[xR_e]/\sigma_e^2$, where $\widehat{\mathbb{E}}^{OS}[xR_e]$ is the Problem Set 1 (2e) out-of-sample forecast made twelve months before t (the forecast of $xR_{w,t}$) and σ_e^2 is the variance of $R_{e,t}$ over the sample (dividing by $T-1$). Sample: December 1940 to December 2021 ($T = 973$), identical in versions (1) and (2), so one set of results is reported.

Method. With $\hat{\mathbf{W}} = \mathbf{I}$, so $\hat{\mathbf{A}} = \hat{\mathbf{G}}'$, $\hat{\gamma}$ solves $\hat{\mathbf{G}}' \widehat{\mathbb{E}}[\boldsymbol{\epsilon}(\hat{\gamma})] = 0$, where $\widehat{\mathbb{E}}[\boldsymbol{\epsilon}(\gamma)]$ is the 2×1 vector of sample moments and $\hat{\mathbf{G}}$ its analytical derivative. Then

$$\begin{aligned} \text{aVar}[\hat{\gamma}] &= \tfrac{1}{T}(\hat{\mathbf{A}}\hat{\mathbf{G}})^{-1}\hat{\mathbf{A}}\hat{\mathbf{S}}\hat{\mathbf{A}}'(\hat{\mathbf{A}}\hat{\mathbf{G}})^{-1\prime}, \\ \text{aVar}\big[\widehat{\mathbb{E}}[\boldsymbol{\epsilon}]\big] &= \tfrac{1}{T}\big[\mathbf{I} - \hat{\mathbf{G}}(\hat{\mathbf{A}}\hat{\mathbf{G}})^{-1}\hat{\mathbf{A}}\big]\hat{\mathbf{S}}\big[\mathbf{I} - \hat{\mathbf{G}}(\hat{\mathbf{A}}\hat{\mathbf{G}})^{-1}\hat{\mathbf{A}}\big]', \end{aligned}$$

with $\hat{\mathbf{S}}$ the Newey–West estimate. Each pricing error has $t_j = \widehat{\mathbb{E}}[\epsilon]_j / \sqrt{\text{aVar}_{jj}}$; the joint test is $\widehat{\mathbb{E}}[\boldsymbol{\epsilon}]' \, \text{aVar}^{+} \, \widehat{\mathbb{E}}[\boldsymbol{\epsilon}] \sim \chi^2(1)$, with the Moore–Penrose pseudo-inverse of the rank-one covariance matrix.

Result (1)–(2). Table 5.

	Estimate	s.e.
$\hat{\gamma}$	3.0130	0.7156
Pricing error	$\widehat{\mathbb{E}}[\epsilon_j]$	t-stat
Market, $xR_{w,t}$	-0.0040	-0.86
Managed portfolio, $z_{t-1} \cdot xR_{w,t}$	0.0019	0.86
$\chi^2(1)$	0.75	
p-value	0.388	
Newey–West lag	21	
Sample	Dec 1940 – Dec 2021 ($T = 973$)	

Table 5: GMM estimate of γ from $\widehat{\mathbb{E}}[R_{w,t}^{-\gamma} \cdot xR_{w,t}] = 0$ and $\widehat{\mathbb{E}}[R_{w,t}^{-\gamma} \cdot (z_{t-1} \cdot xR_{w,t})] = 0$ with an identity weighting matrix ($\hat{\mathbf{A}} = \hat{\mathbf{G}}'$); z_{t-1} as defined in the text. Sample: December 1940 to December 2021 ($T = 973$ months), identical in versions (1) Common and (2) Max. Data: monthly observations of 12-month (annual) real returns from the EQ Dataset. Units: decimals. Standard error next to the estimate and t-statistics: Newey–West (1987, 1994), Bartlett kernel, Newey–West (1994) automatic lag rounded down (selected lag in the table). χ^2: test that the two pricing errors are jointly zero (Moore–Penrose pseudo-inverse).

3(d)

> **Task.** Repeat Question (3c), but this time using a two-step GMM in which the $\hat{\mathbf{S}}^{-1}$ from Question (3c) is used as the weighting matrix when minimizing pricing errors (so that the feasible efficient GMM estimate implicitly uses $\hat{\mathbf{A}} = \hat{\mathbf{G}}'\hat{\mathbf{S}}^{-1}$).

Setup. The moment conditions, data, instrument and sample of Question 3(c), estimated by two-step GMM.

Method. Step 1 is the estimate of Question 3(c) ($\hat{\mathbf{W}} = \mathbf{I}$), and $\hat{\mathbf{S}}_1$ is the Newey–West estimate at that estimate. Step 2 uses $\hat{\mathbf{W}} = \hat{\mathbf{S}}_1^{-1}$: $\hat{\gamma}$ solves $\hat{\mathbf{G}}'\hat{\mathbf{S}}_1^{-1}\,\widehat{\mathbb{E}}[\boldsymbol{\epsilon}(\hat{\gamma})] = 0$, with no further iteration. Standard error, t-statistics and χ^2 use the formulas of Question 3(c) with $\hat{\mathbf{A}} = \hat{\mathbf{G}}'\hat{\mathbf{S}}_1^{-1}$, $\hat{\mathbf{G}}$ at the step-2 estimate, and $\hat{\mathbf{S}}_2$, the Newey–West estimate re-computed at the step-2 estimate.

Result. Table 6.

	Estimate	s.e.
$\hat{\gamma}$ (two-step)	2.9559	0.7058
Pricing error	$\widehat{\mathbb{E}}[\epsilon_j]$	t-stat
Market, $xR_{w,t}$	-0.0022	-0.88
Managed portfolio, $z_{t-1} \cdot xR_{w,t}$	0.0058	0.88
$\chi^2(1)$	0.77	
p-value	0.379	
Newey–West lag, $\hat{S}_1$ / $\hat{S}_2$	21 / 21	
Sample	Dec 1940 – Dec 2021 ($T = 973$)	

Table 6: Two-step GMM estimate of γ from the moment conditions of Question 3(c); $\hat{\mathbf{W}} = \hat{\mathbf{S}}_1^{-1}$ in step 2 and $\hat{\mathbf{A}} = \hat{\mathbf{G}}'\hat{\mathbf{S}}_1^{-1}$ in the sandwich. Sample: December 1940 to December 2021 ($T = 973$ months), identical in versions (1) Common and (2) Max. Data: monthly observations of 12-month (annual) real returns from the EQ Dataset. Units: decimals. Standard error next to the estimate and t-statistics: Newey–West (1987, 1994), Bartlett kernel, Newey–West (1994) automatic lag rounded down, re-selected for $\hat{\mathbf{S}}_1$ and $\hat{\mathbf{S}}_2$ (selected lags in the table). χ^2: test that the two pricing errors are jointly zero (Moore–Penrose pseudo-inverse).

3(e)

> **Task.** Repeat Question (3c), but this time using the moment condition vector $\mathbb{E}[R_{w,t}^{-\gamma} \cdot \mathbf{xR}_t] = \mathbf{0}$, where $\mathbf{xR}$ includes xR_w as well as the excess returns on the 20 decile portfolios in the merge of the BM Dataset and the MOM Dataset.

Setup. GMM estimate of γ from $\mathbb{E}[R_{w,t}^{-\gamma} \cdot xR_{j,t}] = 0$ for $j = 1, \ldots, 21$: the trailing 12-month excess returns of the 20 value-weighted book-to-market and momentum decile portfolios (nominal, over the 12-month compounded Fama–French risk-free rate) and $xR_{w,t}$. Samples: (1) Common: December 1940 to December 2021 ($T = 973$ months); (2) Max: June 1928 to December 2021 ($T = 1123$ months).

Method. As in Question 3(c) with 21 moments: $\hat{\mathbf{W}} = \mathbf{I}$, $\hat{\mathbf{A}} = \hat{\mathbf{G}}'$, the same variance formulas, and the joint test $\widehat{\mathbb{E}}[\boldsymbol{\epsilon}]'\,\mathrm{aVar}^{+}\,\widehat{\mathbb{E}}[\boldsymbol{\epsilon}] \sim \chi^2(20)$ (rank 20).

Result. Table 7.

Panel A: Parameter estimate				
	$\hat{\gamma}$		s.e.	
(1) Common	2.9920		0.7020	
(2) Max	1.9710		0.5830	
Panel B: Pricing errors				
	Book-to-market		Momentum	
Decile	$\widehat{\mathbb{E}}[\epsilon_j]$	t-stat	$\widehat{\mathbb{E}}[\epsilon_j]$	t-stat
(1) Common				
1 (Lo)	−0.0173	−1.52	−0.1048	−7.70
2	0.0014	0.21	−0.0436	−4.92
3	−0.0038	−0.65	−0.0199	−2.30
4	0.0005	0.08	−0.0025	−0.37
5	0.0151	2.32	0.0013	0.18
6	0.0205	2.67	0.0019	0.34
7	0.0058	0.49	0.0092	1.23
8	0.0299	3.02	0.0199	3.84
9	0.0401	3.89	0.0227	3.18
10 (Hi)	0.0335	2.17	0.0500	4.32
Market, $xR_{w,t}$	−0.0033	−0.78		
(2) Max				
1 (Lo)	−0.0102	−1.01	−0.1018	−7.85
2	0.0103	1.44	−0.0436	−5.11
3	0.0021	0.35	−0.0298	−3.27
4	−0.0015	−0.26	−0.0115	−1.65
5	0.0085	1.23	−0.0014	−0.21
6	0.0151	1.83	−0.0002	−0.05
7	−0.0006	−0.07	0.0111	1.58
8	0.0227	2.82	0.0250	4.41
9	0.0333	3.42	0.0299	3.73
10 (Hi)	0.0191	1.32	0.0614	4.69
Market, $xR_{w,t}$	−0.0040	−1.04		
Panel C: Joint test				
	$\chi^2(20)$	p-value	Newey–West lag	T
(1) Common	197.88	<0.001	21	973
(2) Max	185.58	<0.001	23	1123
Sample, (1) Common	Dec 1940 – Dec 2021 ($T = 973$)			
Sample, (2) Max	Jun 1928 – Dec 2021 ($T = 1123$)			

Table 7: GMM estimate of γ from $\widehat{\mathbb{E}}[R_{w,t}^{-\gamma} \cdot xR_{j,t}] = 0$, $j = 1, \ldots, 21$, with an identity weighting matrix ($\hat{\mathbf{A}} = \hat{\mathbf{G}}'$), for both sample versions (rows and blocks labelled by version). Samples: (1) Common: December 1940 to December 2021 ($T = 973$ months); (2) Max: June 1928 to December 2021 ($T = 1123$ months). Data: monthly observations of 12-month returns: the 20 decile portfolios (value-weighted, nominal, over the 12-month compounded risk-free rate) and $xR_{w,t} = e^{r_{e,t}} - e^{r_{f,t}}$ (EQ Dataset, real); $R_{w,t} = e^{r_{e,t}}$. Units: decimals. Panel A: estimate with its standard error in the next column; Panel B: pricing errors with t-statistics, book-to-market and momentum deciles side by side and the market below; Panel C: joint test. Standard error and t-statistics: Newey–West (1987, 1994), Bartlett kernel, Newey–West (1994) automatic lag rounded down (selected lag in Panel C). χ^2: test that the 21 pricing errors are jointly zero (Moore–Penrose pseudo-inverse).

Question 4

Sample periods. Results are reported for two sample versions, both ending in December 2021 (the last EQ observation). Version (1) "Common" uses December 1940 to December 2021 for every subquestion of Questions 3 and 4 (the latest start, set by z in Questions 3(c)–3(d)); version (2) "Max" uses the longest sample available for each subquestion. Version (1) puts Questions 3 and 4 on the same sample, so that differences between the non-linear SDF (Question 3, $M_t = R_{w,t}^{-\gamma}$) and its linearized version (Question 4, $M_t = a - b \cdot R_{w,t}$) are not driven by differences in the sample period; for example, Questions 3(e) and 4(b) use the same 21 moments. Version (2) uses all data available for each subquestion, so each estimate is as precise as the data allow.

Item	(1) Common	(2) Max
Question 3(b)	Dec 1940 – Dec 2021 ($T = 973$)	Dec 1927 – Dec 2021 ($T = 1129$)
Questions 3(c), 3(d)	Dec 1940 – Dec 2021 ($T = 973$)	Dec 1940 – Dec 2021 ($T = 973$)
Question 3(e)	Dec 1940 – Dec 2021 ($T = 973$)	Jun 1928 – Dec 2021 ($T = 1123$)
Questions 4(b)–4(e)	Dec 1940 – Dec 2021 ($T = 973$)	Jun 1928 – Dec 2021 ($T = 1123$)

Questions 3(c) and 3(d) are identical in both versions, so one set of results is reported.

4(a)

> **Task.** [...] (1) Show that we also have (as an approximation) a conditionally linear CAPM, $M_{t+1} \approx a_t - b_t \cdot R_{w,t+1}$, and provide expressions for a_t and b_t. [...]
>
> (2) Highlight a disadvantage of the $M_{t+1} \approx a_t - b_t \cdot R_{w,t+1}$ relative to the original $M_{t+1} = R_{w,t+1}^{-\gamma}$ (beyond the fact that the linear SDF is an approximation).

(1) Conditionally linear CAPM. Let $M_{t+1} = M(F_{t+1}; \theta)$. It can be approximated by a Taylor expansion around $\mathbb{E}_t[F_{t+1}]$:

$$M(F_{t+1};\theta) = \sum_{n=0}^{\infty} \frac{M^{(n)}\big(\mathbb{E}_t[F_{t+1}];\theta\big)}{n!}\big(F_{t+1} - \mathbb{E}_t[F_{t+1}]\big)^n.$$

Terms of second order and above can be neglected. Then,

$$M(F_{t+1};\theta) \approx M\big(\mathbb{E}_t[F_{t+1}];\theta\big) + M'\big(\mathbb{E}_t[F_{t+1}];\theta\big)\big(F_{t+1} - \mathbb{E}_t[F_{t+1}]\big).$$

If we plug in $M(F_{t+1};\theta) = F_{t+1}^{-\gamma}$ with $F_{t+1} = R_{w,t+1}$, then

$$M_{t+1} \approx -\gamma\, \mathbb{E}_t[R_{w,t+1}]^{-\gamma-1} R_{w,t+1} + (\gamma+1)\, \mathbb{E}_t[R_{w,t+1}]^{-\gamma}.$$

Therefore,

$$a_t = (\gamma+1)\, \mathbb{E}_t[R_{w,t+1}]^{-\gamma}, \qquad b_t = \gamma\, \mathbb{E}_t[R_{w,t+1}]^{-\gamma-1}.$$

(2) Disadvantage relative to $M_{t+1} = R_{w,t+1}^{-\gamma}$.

1. The original model, $M_{t+1} = R_{w,t+1}^{-\gamma}$, contains only the factor and the parameter, so there is no concern about additional assumptions or time-varying coefficients.

2. After linearization, a_t and b_t become time-varying, so $\mathbb{E}[xR_j]$ can no longer be written in the form $\mathbb{E}[b_t]' \operatorname{Cov}(F_t, xR_{j,t})$.

3. The time variation in a_t and b_t comes from $\mathbb{E}_t[R_{w,t+1}]$, as shown above. This conditional expectation must be estimated, and the choice of how to estimate it (e.g., with a linear model or some other method) introduces estimation error. The model could be wrong, or there could be estimation error even if the model is correct.

4. By imposing additional assumptions or a model on a_t and b_t, they can be made time-invariant. However, if that model is wrong or the assumptions are inconsistent with reality, the estimates move further away from reality.

5. $\mathbb{E}_t[\mathbf{xR}_{t+1}] = b_t' \operatorname{Cov}_t(F_{t+1}, \mathbf{xR}_{t+1})$ holds, but from the perspective of the unconditional implications used in GMM, it can be written as $\mathbb{E}[xR_j] = \mathbb{E}\big[b_t' \operatorname{Cov}_t(F_{t+1}, xR_{j,t+1})\big]$.

6. But again, b_t is determined by $\mathbb{E}_t[R_{w,t}]$. As in Problem Set 1, we already checked that estimating the equity premium *very well*, either in sample (IS) or out of sample (OS), is really difficult.

4(b)

Task. (1) For simplicity, let's consider the SDF model $M_t = a - b \cdot R_{w,t}$ with the normalization $\mathbb{E}[M] = 1$, which can be written as $M_t = 1 - b \cdot (R_{w,t} - \mathbb{E}[R_w])$. Estimate b by GMM with the unconditional SDF equation $\mathbb{E}[M \cdot \mathbf{xR}_t] = \mathbf{0}$, where $\mathbf{xR}$ includes xR_w as well as the excess returns on the 20 decile portfolios in the merge of the BM Dataset and the MOM Dataset. Use an identity weighting matrix when minimizing pricing errors (so that the GMM estimate implicitly uses $\hat{\mathbf{A}} = \hat{\mathbf{G}}'$).

(2) Produce a table that reports the b estimate as well as its standard error. The table should also report the value and t-statistic for the pricing error of each moment condition as well as the p-value for a χ^2 test with the null hypothesis that the pricing errors are jointly zero. When estimating $\operatorname{aVar}[\hat{\gamma}]$ and $\operatorname{aVar}[\widehat{\mathbb{E}}[\epsilon(\hat{\gamma})]]$, estimate S using Newey and West (1987, 1994).

Setup. SDF $M_t = 1 - b\,(R_{w,t} - \mu)$ with $\mu = \mathbb{E}[R_{w,t}]$ (normalization $\mathbb{E}[M] = 1$). Test assets: the 21 excess returns of Question 3(e) ($xR_{j,t}$, $j = 1, \ldots, 20$, and $xR_{w,t}$), with $R_{w,t} = e^{r_{e,t}}$. Samples: (1) Common: December 1940 to December 2021 ($T = 973$ months); (2) Max: June 1928 to December 2021 ($T = 1123$ months).

Method. The parameters $\boldsymbol{\theta} = (b, \mu)$ satisfy 22 moments: the pricing moments $\mathbb{E}[M_t \cdot xR_{j,t}] = 0$, $j = 1, \ldots, 21$, and $\mathbb{E}[R_{w,t} - \mu] = 0$. The last one is set exactly to zero, so $\hat{\mu} = \overline{R}_w$, and b uses identity weights on the pricing moments:

$$\hat{b} = (\hat{\mathbf{d}}'\hat{\mathbf{d}})^{-1}\hat{\mathbf{d}}'\,\widehat{\mathbb{E}}[\mathbf{xR}], \qquad \hat{\mathbf{d}} = \widehat{\operatorname{Cov}}[R_{w,t}, \mathbf{xR}_t] \text{ (dividing by } T).$$

This is GMM with $\hat{\mathbf{A}} = \begin{bmatrix} \hat{\mathbf{G}}_b' & 0 \\ \mathbf{0}' & 1 \end{bmatrix}$, where $\hat{\mathbf{G}}_b$ is the derivative of the pricing moments with respect to b (rows: parameters b and μ; columns: the 21 pricing moments and the μ moment). Standard errors, t-statistics and the joint test use the formulas of Question 3(c) with this $\hat{\mathbf{A}}$, the 22×22 Newey–West $\hat{\mathbf{S}}$, and the 21×21 pricing block of $\operatorname{aVar}\big[\widehat{\mathbb{E}}[\boldsymbol{\epsilon}]\big]$ (rank 20), so they account for estimating μ.

Result (1)–(2). Table 8.

Panel A: Parameter estimate

	$\hat{b}$	s.e.
(1) Common	3.2860	0.7797
(2) Max	1.9751	0.6047

Panel B: Pricing errors

	Book-to-market		Momentum	
Decile	$\widehat{\mathbb{E}}[\epsilon_j]$	t-stat	$\widehat{\mathbb{E}}[\epsilon_j]$	t-stat
(1) Common				
1 (Lo)	−0.0170	−1.38	−0.1164	−9.05
2	0.0017	0.23	−0.0459	−4.97
3	−0.0029	−0.46	−0.0220	−2.47
4	0.0015	0.23	−0.0046	−0.64
5	0.0143	2.03	0.0011	0.15
6	0.0209	2.80	0.0014	0.25
7	0.0068	0.64	0.0087	1.16
8	0.0285	2.96	0.0213	3.83
9	0.0375	3.56	0.0261	3.25
10 (Hi)	0.0273	1.80	0.0553	4.39
Market, $xR_{w,t}$	−0.0034	−0.77		
(2) Max				
1 (Lo)	−0.0062	−0.56	−0.1110	−8.76
2	0.0163	1.98	−0.0493	−5.45
3	0.0054	0.80	−0.0275	−3.00
4	−0.0017	−0.28	−0.0128	−1.84
5	0.0109	1.60	0.0000	0.00
6	0.0163	2.12	0.0003	0.06
7	−0.0014	−0.15	0.0163	2.16
8	0.0195	2.34	0.0298	4.57
9	0.0266	2.56	0.0360	4.02
10 (Hi)	0.0083	0.56	0.0709	5.10
Market, $xR_{w,t}$	−0.0002	−0.03		

Panel C: Joint test

	$\chi^2(20)$	p-value	Newey–West lag	T
(1) Common	304.77	<0.001	22	973
(2) Max	229.37	<0.001	22	1123
Sample, (1) Common	Dec 1940 – Dec 2021 ($T = 973$)			
Sample, (2) Max	Jun 1928 – Dec 2021 ($T = 1123$)			

Table 8: First-stage GMM estimate of b in $M_t = 1 - b\,(R_{w,t} - \mu)$ with identity weights on the 21 pricing moments and $\mathbb{E}[R_{w,t} - \mu] = 0$ satisfied exactly, for both sample versions (rows and blocks labelled by version). Samples: (1) Common: December 1940 to December 2021 ($T = 973$ months); (2) Max: June 1928 to December 2021 ($T = 1123$ months). Data: monthly observations of 12-month returns: the 20 decile portfolios (value-weighted, nominal, over the 12-month compounded risk-free rate) and $xR_{w,t} = e^{r_{e,t}} - e^{r_{f,t}}$ (EQ Dataset, real); $R_{w,t} = e^{r_{e,t}}$. Units: decimals. Panel A: estimate with its standard error in the next column; Panel B: pricing errors with t-statistics, book-to-market and momentum deciles side by side and the market below; Panel C: joint test. Standard error and t-statistics: Newey–West (1987, 1994), Bartlett kernel, Newey–West (1994) automatic lag rounded down (selected lag in Panel C). χ^2: test that the 21 pricing errors are jointly zero (Moore–Penrose pseudo-inverse).

4(c)

> **Task.** Repeat Question (4b), but this time using the efficient GMM. Note that in this case you have an analytical expression for the GMM estimate of b and can easily compute it (without solving any numerical optimization problem).

Setup. The model, data and sample of Question 4(b), estimated by efficient GMM.

Method. With $\hat{\mu} = \overline{R}_w$, for any weighting matrix $\hat{\mathbf{W}}$ on the 21 pricing moments b has the closed form

$$\hat{b}(\hat{\mathbf{W}}) = (\hat{\mathbf{d}}'\hat{\mathbf{W}}\hat{\mathbf{d}})^{-1}\hat{\mathbf{d}}'\hat{\mathbf{W}}\,\widehat{\mathbb{E}}[\mathbf{xR}],$$

so no numerical optimization is needed. Two-step: $\hat{\mathbf{W}} = \hat{\mathbf{S}}^{-1}$, with $\hat{\mathbf{S}}$ the 21×21 pricing block of the Newey–West estimate at the estimate of Question 4(b). Iterated: $\hat{\mathbf{S}}$ and $\hat{b}$ are updated until $|\hat{b}_{h+1} - \hat{b}_h| < 10^{-10}$. Standard errors, t-statistics and χ^2 use the formulas of Question 4(b) with $\hat{\mathbf{A}} = \begin{bmatrix} \hat{\mathbf{G}}_b'\hat{\mathbf{W}} & 0 \\ \mathbf{0}' & 1 \end{bmatrix}$ and $\hat{\mathbf{S}}$ re-estimated at each estimate.

Result. Tables 9 ((1) Common) and 10 ((2) Max).

Panel A: Parameter estimates, (1) Common

	$\hat{b}$	s.e.
4(b), $\hat{\mathbf{W}} = \mathbf{I}$	3.2860	0.7797
Two-step	6.8884	0.8574
Iterated	5.7470	0.7512

Panel B: Pricing errors, (1) Common

	Book-to-market		Momentum	
Decile	$\widehat{\mathbb{E}}[\epsilon_j]$	t-stat	$\widehat{\mathbb{E}}[\epsilon_j]$	t-stat
4(b), $\hat{\mathbf{W}} = \mathbf{I}$				
1 (Lo)	−0.0170	−1.38	−0.1164	−9.05
2	0.0017	0.23	−0.0459	−4.97
3	−0.0029	−0.46	−0.0220	−2.47
4	0.0015	0.23	−0.0046	−0.64
5	0.0143	2.03	0.0011	0.15
6	0.0209	2.80	0.0014	0.25
7	0.0068	0.64	0.0087	1.16
8	0.0285	2.96	0.0213	3.83
9	0.0375	3.56	0.0261	3.25
10 (Hi)	0.0273	1.80	0.0553	4.39
Market, $xR_{w,t}$	−0.0034	−0.77		
Two-step				
1 (Lo)	−0.1275	−5.01	−0.2669	−7.52
2	−0.1008	−5.05	−0.1621	−5.31
3	−0.1012	−5.08	−0.1293	−5.38
4	−0.0954	−4.58	−0.1047	−4.66
5	−0.0822	−4.01	−0.0916	−5.41
6	−0.0760	−3.53	−0.0986	−5.11
7	−0.0926	−3.28	−0.0874	−5.03
8	−0.0783	−3.37	−0.0807	−4.11
9	−0.0738	−3.25	−0.0812	−3.44
10 (Hi)	−0.1030	−3.62	−0.0705	−2.56
Market, $xR_{w,t}$	−0.1031	−5.13		
Iterated				
1 (Lo)	−0.0925	−4.18	−0.2192	−7.63
2	−0.0683	−4.22	−0.1253	−4.91
3	−0.0701	−3.96	−0.0953	−4.87
4	−0.0647	−3.42	−0.0730	−3.95
5	−0.0516	−2.73	−0.0623	−4.08
6	−0.0453	−2.25	−0.0669	−3.95
7	−0.0611	−2.46	−0.0569	−3.63
8	−0.0445	−2.08	−0.0484	−2.69
9	−0.0385	−1.80	−0.0472	−2.20
10 (Hi)	−0.0617	−2.48	−0.0306	−1.17
Market, $xR_{w,t}$	−0.0715	−4.23		

Panel C: Joint test and Newey–West lags, (1) Common

	$\chi^2(20)$	p-value	Lag, $\hat{\mathbf{W}}$ / sandwich	Iterations
4(b), $\hat{\mathbf{W}} = \mathbf{I}$	304.77	<0.001	– / 22	–
Two-step	181.59	<0.001	22 / 20	–
Iterated	254.51	<0.001	21 / 21	20
Sample	Dec 1940 – Dec 2021 ($T = 973$)			

Table 9: Efficient GMM estimates of b in $M_t = 1 - b\,(R_{w,t} - \mu)$: Question 4(b) ($\hat{\mathbf{W}} = \mathbf{I}$, Table 8, for reference), two-step, and iterated; sample version (1) Common. Sample: December 1940 to December 2021 ($T = 973$ months). Data: monthly observations of 12-month returns, as in Table 8. Units: decimals. Panel A: estimates with their standard errors in the next column; Panel B: pricing errors with t-statistics for each estimate, book-to-market and momentum deciles side by side and the market below; Panel C: joint tests, Newey–West lags and iterations. Standard errors and t-statistics: Newey–West (1987, 1994), Bartlett kernel, Newey–West (1994) automatic lag rounded down and re-selected each time $\hat{\mathbf{S}}$ is estimated (lags of the weighting matrix and of the sandwich in Panel C). χ^2: test that the 21 pricing errors are jointly zero (Moore–Penrose pseudo-inverse). "–": not applicable.

Panel A: Parameter estimates, (2) Max

	$\hat{b}$	s.e.
4(b), $\hat{\mathbf{W}} = \mathbf{I}$	1.9751	0.6047
Two-step	3.9531	0.6284
Iterated	4.1161	0.6231

Panel B: Pricing errors, (2) Max

	Book-to-market		Momentum	
Decile	$\widehat{\mathbb{E}}[\epsilon_j]$	t-stat	$\widehat{\mathbb{E}}[\epsilon_j]$	t-stat
4(b), $\hat{\mathbf{W}} = \mathbf{I}$				
1 (Lo)	−0.0062	−0.56	−0.1110	−8.76
2	0.0163	1.98	−0.0493	−5.45
3	0.0054	0.80	−0.0275	−3.00
4	−0.0017	−0.28	−0.0128	−1.84
5	0.0109	1.60	0.0000	0.00
6	0.0163	2.12	0.0003	0.06
7	−0.0014	−0.15	0.0163	2.16
8	0.0195	2.34	0.0298	4.57
9	0.0266	2.56	0.0360	4.02
10 (Hi)	0.0083	0.56	0.0709	5.10
Market, $xR_{w,t}$	−0.0002	−0.03		
Two-step				
1 (Lo)	−0.0926	−3.41	−0.2343	−5.57
2	−0.0614	−2.99	−0.1581	−3.85
3	−0.0751	−3.44	−0.1185	−3.41
4	−0.0932	−2.80	−0.1069	−2.91
5	−0.0750	−2.38	−0.0838	−2.89
6	−0.0744	−2.08	−0.0883	−2.81
7	−0.0959	−2.46	−0.0640	−2.51
8	−0.0847	−2.28	−0.0531	−2.28
9	−0.0846	−2.20	−0.0502	−2.14
10 (Hi)	−0.1212	−2.54	−0.0210	−0.90
Market, $xR_{w,t}$	−0.0833	−3.10		
Iterated				
1 (Lo)	−0.0998	−3.23	−0.2445	−4.75
2	−0.0678	−3.07	−0.1670	−3.25
3	−0.0817	−2.98	−0.1260	−3.13
4	−0.1008	−2.41	−0.1146	−2.54
5	−0.0821	−2.17	−0.0907	−2.51
6	−0.0818	−1.90	−0.0956	−2.47
7	−0.1037	−2.19	−0.0706	−2.43
8	−0.0933	−1.96	−0.0599	−2.14
9	−0.0937	−1.83	−0.0573	−2.09
10 (Hi)	−0.1319	−2.10	−0.0286	−1.16
Market, $xR_{w,t}$	−0.0901	−2.86		

Panel C: Joint test and Newey–West lags, (2) Max

	$\chi^2(20)$	p-value	Lag, $\hat{\mathbf{W}}$ / sandwich	Iterations
4(b), $\hat{\mathbf{W}} = \mathbf{I}$	229.37	<0.001	– / 22	–
Two-step	132.44	<0.001	22 / 20	–
Iterated	128.64	<0.001	20 / 20	32
Sample	Jun 1928 – Dec 2021 ($T = 1123$)			

Table 10: Efficient GMM estimates of b in $M_t = 1 - b\,(R_{w,t} - \mu)$: Question 4(b) ($\hat{\mathbf{W}} = \mathbf{I}$, Table 8, for reference), two-step, and iterated; sample version (2) Max. Sample: June 1928 to December 2021 ($T = 1123$ months). Data: monthly observations of 12-month returns, as in Table 8. Units: decimals. Panel A: estimates with their standard errors in the next column; Panel B: pricing errors with t-statistics for each estimate, book-to-market and momentum deciles side by side and the market below; Panel C: joint tests, Newey–West lags and iterations. Standard errors and t-statistics: Newey–West (1987, 1994), Bartlett kernel, Newey–West (1994) automatic lag rounded down and re-selected each time $\hat{\mathbf{S}}$ is estimated (lags of the weighting matrix and of the sandwich in Panel C). χ^2: test that the 21 pricing errors are jointly zero (Moore–Penrose pseudo-inverse). "–": not applicable.

4(d)

Task. (1) Estimate λ and α_j for each asset in Question (4b) using time series regressions for the linear beta pricing model implied by the $M_t = a - b \cdot R_{w,t}$ SDF.

(2) Produce a table that reports the λ estimate as well as its standard error (using Newey and West (1987, 1994) on a time series regression of $R_{w,t}$ on a constant). The table should also report the value and t-statistic for each α_j (using Newey and West (1987, 1994) on the respective time series factor regression).

(3) Finally, the table should also report the p-value for the GRS test with the null hypothesis that the pricing errors are jointly zero.

Setup. Time-series estimates of the linear beta pricing model implied by $M_t = a - b \cdot R_{w,t}$. With $R_{f,t}$ treated as constant, $M_t = (a - bR_f) - b \cdot xR_{w,t}$, so the factor is the excess return $F_t = xR_{w,t}$. Test assets as in Question 4(b). Samples: (1) Common: December 1940 to December 2021 ($T = 973$ months); (2) Max: June 1928 to December 2021 ($T = 1123$ months).

Method. $\hat{\lambda} = \widehat{\mathbb{E}}[F]$, with the Newey–West standard error from a time-series regression of F_t on a constant. For each asset, $\hat{\alpha}_j$ is the intercept of the time-series regression $xR_{j,t} = \alpha_j + \beta_j F_t + \varepsilon_{j,t}$, with a Newey–West t-statistic. The GRS test (Gibbons, Ross and Shanken, 1989) for the 20 decile portfolios is

$$\text{GRS} = \frac{T - J - K}{J}\left(1 + \bar{F}'\hat{\Sigma}_F^{-1}\bar{F}\right)^{-1}\hat{\boldsymbol{\alpha}}'\hat{\boldsymbol{\Sigma}}_\varepsilon^{-1}\hat{\boldsymbol{\alpha}} \sim F(J, T - J - K),$$

with $J = 20$, $K = 1$, $\hat{\boldsymbol{\Sigma}}_\varepsilon$ dividing by $T - 2$ and $\hat{\Sigma}_F$ by $T - 1$.

Result (1)–(3). Table 11.

Panel A: Factor risk premium		
	$\hat{\lambda}$	s.e.
(1) Common	0.0874	0.0159
(2) Max	0.0828	0.0186

Panel B: Pricing errors	Book-to-market		Momentum	
Decile	$\hat{\alpha}_j$	t-stat	$\hat{\alpha}_j$	t-stat
(1) Common				
1 (Lo)	−0.0122	−1.38	−0.1157	−8.49
2	0.0059	1.15	−0.0424	−3.59
3	0.0011	0.15	−0.0186	−2.28
4	0.0043	0.50	−0.0017	−0.17
5	0.0163	1.65	0.0035	0.49
6	0.0239	2.25	0.0044	0.80
7	0.0089	0.66	0.0118	1.69
8	0.0308	2.33	0.0245	3.87
9	0.0395	2.68	0.0294	3.62
10 (Hi)	0.0297	1.52	0.0597	5.02
Market, $xR_{w,t}$	0.0000	–		
(2) Max				
1 (Lo)	−0.0052	−0.64	−0.1121	−7.55
2	0.0171	2.90	−0.0487	−4.09
3	0.0064	0.97	−0.0279	−3.25
4	−0.0017	−0.21	−0.0129	−1.42
5	0.0104	1.20	0.0003	0.04
6	0.0158	1.60	0.0007	0.13
7	−0.0018	−0.15	0.0161	2.59
8	0.0193	1.57	0.0298	4.82
9	0.0272	1.76	0.0361	4.20
10 (Hi)	0.0086	0.42	0.0712	5.51
Market, $xR_{w,t}$	0.0000	–		

Panel C: GRS test and Newey–West lags	GRS	p-value	Lag, $\hat{\lambda}$	Lags, $\hat{\alpha}_j$
(1) Common, $F(20, 952)$	50.16	<0.001	22	21–23
(2) Max, $F(20, 1102)$	49.65	<0.001	23	19–23
Sample, (1) Common	Dec 1940 – Dec 2021 ($T = 973$)			
Sample, (2) Max	Jun 1928 – Dec 2021 ($T = 1123$)			

Table 11: Time-series estimates of λ and α_j with the factor $F_t = xR_{w,t}$, for both sample versions (rows and blocks labelled by version). Samples: (1) Common: December 1940 to December 2021 ($T = 973$ months); (2) Max: June 1928 to December 2021 ($T = 1123$ months). Data: monthly observations of 12-month returns, as in Table 8. Units: decimals. Panel A: $\hat{\lambda}$ with its standard error in the next column; Panel B: $\hat{\alpha}_j$ with t-statistics, book-to-market and momentum deciles side by side and the market below; Panel C: GRS test and Newey–West lags. Standard error of $\hat{\lambda}$ and t-statistics of $\hat{\alpha}_j$: Newey–West (1987, 1994), Bartlett kernel, Newey–West (1994) automatic lag rounded down, selected for each regression (lags in Panel C). GRS: $F(20, 1102)$ test that the 20 decile-portfolio α_j are jointly zero. "–": undefined (for the factor itself $\hat{\alpha} = 0$ by construction).

4(e)

> **Task.** (1) Estimate λ and α_j for each asset in Question (4b) using Fama-MacBeth regressions for the linear beta pricing model implied by the $M_t = a - b \cdot R_{w,t}$ SDF.
>
> (2) Produce a table that reports the λ estimate as well as its standard error. The table should also report the value and t-statistic for each α_j. The standard error for $\hat{\lambda}$ ($\hat{\alpha}_j$) should come from a time series regression of $\hat{\lambda}_t$ ($\hat{\alpha}_{j,t}$) on a constant with Newey and West (1987, 1994) standard errors.

Setup. Fama–MacBeth estimates of the linear beta pricing model implied by $M_t = a - b \cdot R_{w,t}$, with the factor $F_t = R_{w,t}$ (the gross return; no constant-R_f assumption). Test assets as in Question 4(b). Samples: (1) Common: December 1940 to December 2021 ($T = 973$ months); (2) Max: June 1928 to December 2021 ($T = 1123$ months).

Method. First pass: for each asset, the full-sample time-series regression $xR_{j,t} = \beta_{0,j} + \beta_j R_{w,t} + \varepsilon_{j,t}$ gives $\hat{\beta}_j$. Second pass: for each month, the cross-sectional regression without intercept $xR_{j,t} = \lambda_t \hat{\beta}_j + \alpha_{j,t}$, $j = 1, \ldots, 21$, gives $\hat{\lambda}_t$ and $\hat{\alpha}_{j,t}$. Then $\hat{\lambda}$ and $\hat{\alpha}_j$ are the time averages, with Newey–West standard errors from a time-series regression of $\hat{\lambda}_t$ ($\hat{\alpha}_{j,t}$) on a constant.

Result (1)–(2). Table 12.

Panel A: Factor risk premium

	$\hat{\lambda}$	s.e.
(1) Common	0.0947	0.0163
(2) Max	0.0832	0.0179

Panel B: Pricing errors

	Book-to-market		Momentum	
Decile	$\hat{\alpha}_j$	t-stat	$\hat{\alpha}_j$	t-stat
(1) Common				
1 (Lo)	−0.0170	−1.52	−0.1164	−7.76
2	0.0017	0.23	−0.0459	−5.52
3	−0.0029	−0.46	−0.0220	−2.83
4	0.0015	0.24	−0.0046	−0.68
5	0.0143	2.22	0.0011	0.15
6	0.0209	3.59	0.0014	0.27
7	0.0068	0.84	0.0087	1.36
8	0.0285	3.20	0.0213	3.32
9	0.0375	4.11	0.0261	3.29
10 (Hi)	0.0273	2.03	0.0553	4.53
Market, $xR_{w,t}$	−0.0034	−0.90		
(2) Max				
1 (Lo)	−0.0062	−0.59	−0.1110	−7.76
2	0.0163	1.91	−0.0493	−5.42
3	0.0054	0.89	−0.0275	−3.12
4	−0.0017	−0.25	−0.0128	−1.88
5	0.0109	1.59	0.0000	0.00
6	0.0163	2.55	0.0003	0.06
7	−0.0014	−0.18	0.0163	2.11
8	0.0195	2.36	0.0298	4.35
9	0.0266	2.77	0.0360	4.24
10 (Hi)	0.0083	0.57	0.0709	5.24
Market, $xR_{w,t}$	−0.0002	−0.03		

Panel C: Newey–West lags

	Lag, $\hat{\lambda}$	Lags, $\hat{\alpha}_j$	T
(1) Common	22	21–23	973
(2) Max	23	22–23	1123
Sample, (1) Common	Dec 1940 – Dec 2021 ($T = 973$)		
Sample, (2) Max	Jun 1928 – Dec 2021 ($T = 1123$)		

Table 12: Fama–MacBeth estimates of λ and α_j with the factor $F_t = R_{w,t}$, for both sample versions (rows and blocks labelled by version). Samples: (1) Common: December 1940 to December 2021 ($T = 973$ months); (2) Max: June 1928 to December 2021 ($T = 1123$ months). Data: monthly observations of 12-month returns, as in Table 8. Units: decimals. Panel A: $\hat{\lambda}$ with its standard error in the next column; Panel B: $\hat{\alpha}_j$ with t-statistics, book-to-market and momentum deciles side by side and the market below; Panel C: Newey–West lags. Standard errors and t-statistics: Newey–West (1987, 1994) on a time-series regression of $\hat{\lambda}_t$ ($\hat{\alpha}_{j,t}$) on a constant, Bartlett kernel, Newey–West (1994) automatic lag rounded down, selected for each series (lags in Panel C).